

Introduction to Machine Learning — Study Notes

1. What is Machine Learning?

Machine learning (ML) is a branch of artificial intelligence that enables systems to learn from data and improve their performance on tasks without being explicitly programmed. Instead of writing rules manually, ML algorithms identify patterns in data and use them to make predictions or decisions.

2. Types of Machine Learning

There are three primary types of machine learning: Supervised Learning, Unsupervised Learning, and Reinforcement Learning.

Supervised Learning uses labeled training data where both inputs and desired outputs are provided. Common algorithms include Linear Regression, Logistic Regression, Decision Trees, Random Forests, and Support Vector Machines (SVM).

Unsupervised Learning works with unlabeled data to discover hidden patterns or groupings. Key techniques include K-Means Clustering, Hierarchical Clustering, and Principal Component Analysis (PCA) for dimensionality reduction.

Reinforcement Learning trains agents to make decisions by rewarding correct actions and penalizing wrong ones. It is widely used in robotics, game playing (like AlphaGo), and autonomous vehicles.

3. Bias-Variance Tradeoff

The bias-variance tradeoff is a fundamental concept in ML. Bias refers to errors from incorrect assumptions in the model — a high-bias model underfits the data. Variance refers to sensitivity to small fluctuations in the training data — a high-variance model overfits.

The goal is to find the sweet spot that minimizes total error. Techniques like cross-validation, regularization (L1/Lasso, L2/Ridge), and ensemble methods help manage this tradeoff.

4. Neural Networks and Deep Learning

Neural networks are computing systems inspired by biological brains. A basic neural network consists of an input layer, one or more hidden layers, and an output layer. Each connection has a weight that is adjusted during training using backpropagation and gradient descent.

Deep learning refers to neural networks with many hidden layers. Convolutional Neural Networks (CNNs) excel at image recognition. Recurrent Neural Networks (RNNs) and Transformers are used for sequential data like text and speech.

5. Model Evaluation Metrics

For classification tasks, common metrics include Accuracy, Precision, Recall, F1-Score, and the ROC-AUC curve. For regression tasks, common metrics include Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2).

Confusion matrices are useful for understanding true positives, false positives, true negatives, and false negatives in classification problems.

6. Feature Engineering

Feature engineering is the process of transforming raw data into meaningful inputs for ML models. Techniques include normalization and standardization, one-hot encoding for categorical variables, handling missing values through imputation, and creating interaction features.

7. Popular ML Libraries

Scikit-learn is the most widely used Python library for classical ML algorithms. TensorFlow and PyTorch are the leading deep learning frameworks. Pandas and NumPy are essential for data manipulation. Matplotlib and Seaborn are used for data visualization.

8. Real-World Applications

Machine learning is applied across many domains: healthcare (disease diagnosis, drug discovery), finance (fraud detection, algorithmic trading), natural language processing (chatbots, translation, sentiment analysis), computer vision (facial recognition, autonomous driving), and recommendation systems (Netflix, Amazon, Spotify).